On the Danger of Outsourcing the Struggle

On thinking, tools, and those rare moments of recognition

Lee McCulloch-James London, 2026

Before the Paint Goes On

Dear Student of Mathematics and Physics,

Before we talk about thinking and tools, let me offer a simple metaphor from something far more ordinary than mathematics or physics: painting an external door.

When a door is painted properly, the glossy surface you see at the end is the least important part of the process. What matters is everything that happens before the gloss ever touches the door.

If the door has been painted before, the first step is not painting at all. It is *sanding*. The old surface must be roughened so that the new layer has something to grip. Painters call this giving the surface a *key*. Without that key, the new paint will sit beautifully for a while and then peel away.

After sanding comes the **primer**. The primer seals the surface and creates the foundation to which everything else adheres. This stage is slow and unglamorous, but it determines whether the paint will last for years or fail within months.

Then comes the **undercoat**. This is where the real colour begins to take shape. It is not yet the finished surface, but the form is now there. The undercoat carries the weight of the paint system. It is the layer that gives the finish its strength.

Only at the very end does the **gloss** go on. The gloss gives the surface its lustre. It is what people notice first. But the gloss is not what makes the paint durable. The durability lives underneath.

Thinking works in much the same way. Old ideas sometimes need sanding so that new understanding can attach to them. The primer stage is where the real intellectual work happens: the slow process of building the conceptual surface on which understanding can adhere. The undercoat is your own reasoning as it takes shape after hours of rumination and false starts. Only then should the gloss appear.

Large language models are very good at producing gloss. They can even help shape an undercoat once the structure already exists. But they cannot do the sanding, and they cannot lay the primer. Those layers must always be yours. If they are not, the finish may look convincing, but it will not last.

With matters allegorical off my chest, let us return to the substance of this letter.

The Old Loops

In mathematics, the cognitive process has always looked something like this:

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intuition → internal dialogue → clarity
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You notice something. You argue with yourself about it. You write it down. You cross it out. You write it again. Slowly, painfully, the idea takes shape. The internal dialogue is where the work happens. It is slow because understanding is slow. You are not just producing sentences. You are testing whether you believe what the sentences say.

In physics, there is a parallel loop, but it passes through the world:

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observation → model → prediction → test
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You see something that does not fit your expectations. You build a model to explain it. The model makes a prediction. You test the prediction against reality. If the prediction fails, you revise the model. If it holds, you have learned something. The test is not optional. It is what separates physics from storytelling.

Both loops require a period of struggle. In mathematics, the struggle is internal: you sit with the algebra until the structure reveals itself. In physics, the struggle is also external: the universe is under no obligation to confirm your model, and the moment of failure, the experiment that does not go as predicted, is often more instructive than the moment of success.

The New Loops

Large language models offer a substitution in both disciplines:

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intuition → dialogue with tool → clarity
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observation → dialogue with tool → plausible model
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The internal dialogue is replaced by an external one. You have a fragmentary idea. You describe it to the machine. The machine returns a candidate articulation: polished, structured, plausible. You read it and think, "Yes, that is what I meant."

In physics the danger has a specific shape. The machine can produce a model that is coherent, consistent with known results, and expressed in correct formalism. What it cannot do is test that model against reality. It has no laboratory. It has no apparatus. It cannot run the experiment. And a model that has not been tested against the world is not physics. It is a hypothesis dressed in the language of physics.

This is seductive in both disciplines. It is also, for someone at your stage, genuinely dangerous.

What the Machine Does and Does Not Do

The machine generates candidate structures. It is very good at this. Given fragments, it produces coherent wholes. Given a question, it produces a plausible answer. Given half a proof, it produces the other half in fluent mathematical prose. Given a set of experimental observations, it produces a model that sounds reasonable.

What it does not do is *understand whether the structure is correct.* It does not know if the proof is valid. It does not know if the result is interesting. It does not know if the argument, however polished, has a hole in its centre. It does not know if the model contradicts an experiment performed in 1987 that it was not trained on. It is generating text that is statistically consistent with the corpus of scientific and mathematical exposition it has seen. Sometimes that text is brilliant. Sometimes it is fluent nonsense. The machine cannot tell the difference. You must.

A Historical Perspective

Euler had human computers who performed calculations for him. Gauss had tables of primes. Ramanujan tested his conjectures by hand-computing hundreds of terms. None of them were diminished by the mechanical step. They were defined by what happened after it: the recognition of structure, the judgment of significance, the decision about what deserved to become a theorem.

In physics the same is true. Tycho Brahe spent decades collecting positional data on Mars. He did not understand it. Kepler took Brahe's data and, after years of failed models, recognised that the orbits were ellipses. The data was the mechanical step. The recognition was the science. Later, Eddington sailed to an island off West Africa to photograph a solar eclipse and test whether starlight bent as Einstein predicted. He could have stayed home and declared the theory beautiful. He did not. He went and measured. That is the difference between physics and plausible narrative.

Recognition as the Source of Authority

When I look at a candidate structure produced during dialogue with an LLM tool and say, "that is wrong," or "that is right but uninteresting," or "that is the point," I am performing the same act of judgment I would perform if the candidate nascent idea had surfaced from my own thinking at three in the morning.

But here is the crucial difference between you and me. I have spent thirty years building an internal machinery that lets me make those judgment calls, pretty much. I have assimilated and accommodated, in Piaget's language, a body of mathematical and physical knowledge. I have made enough mistakes, followed enough dead ends, run the course of a nascent thought to some logical conclusion that gave the wrong answer, and sat with enough confusion, that I have a sense of when an argument is actually load-bearing and when it merely appears load-bearing. That sense is hard-won. It cannot be shortcut.

The danger is this. If you outsource the internal dialogue before you have built the internal machinery, you lose the opportunity to gain the capacity to perform that crucial recognition step. The LLM tool gives you clarity before you have earned it. The result looks like understanding but it is not understanding. It is merely glossy fluency without a well-primed foundation. And you will not know the difference until the moment someone asks you to defend what you have written, and you find that you cannot.

How do you prove an idea is yours? In the film *Working Girl*, Tess McGill is asked exactly this question. Her boss has stolen her idea. In front of a room full of executives, Tess is challenged to demonstrate that the idea originated with her. She does it the only way that works: she traces the chain of thought. She shows the sequence of newspaper clippings, the connections she made, the path from one fragment to the next. The idea is hers because she can reconstruct, step by step, how she arrived at it.

Working Girl (1988). Tess proves ownership of an idea by tracing the chain of thought that produced it. Video reference: `https://www.youtube.com/embed/m8WRHX2t_hE`

You should be able to do this with every piece of work you submit. Not because someone might steal your idea, but because if you cannot trace the path from observation to conclusion, you do not own the conclusion. You are merely holding it. And if a machine generated the intermediate steps, the path is not yours to trace.

Piaget's Ladder

On Assimilation and Accommodation

Piaget described two processes by which we build knowledge. **Assimilation** is fitting new information into existing mental structures. **Accommodation** is restructuring those mental frameworks when the new information does not fit. The second process is the hard one. It requires discomfort. It requires sitting with the fact that your current understanding is inadequate and needs to be rebuilt.

A large language model will never make you accommodate. It will produce a smooth, coherent account that you can assimilate immediately, because it is designed to produce text that reads as already understood. The struggle, the confusion, the rebuilding: those are precisely the experiences that the tool eliminates. And they are precisely the experiences that build the machinery of recognition.

Every mathematician or physicist has a story about a period of confusion that preceded their deepest work. The confusion was not a failure of the process. It was the process. The period in which nothing made sense was the period in which their mental structures were being rebuilt to accommodate something genuinely new.

In mathematics, accommodation looks like the moment when you realise that your intuition about infinity is wrong and you have to rebuild your concept of number. For example, the discovery that the rational numbers between 0 and 1 are not merely many but infinitely many,

and can even be listed one by one, forces you to rethink what infinity means. Or the moment when a proof you believed for a week turns out to have a gap and you have to start again.

In physics, accommodation looks like the moment when you drop two objects of different mass and they hit the ground at the same time, and your Aristotelian expectation is shattered. Or the moment when you, as Max Planck, calculate the ultraviolet catastrophe from classical thermodynamics and realise the theory must be wrong, that energy must be quantised, and that the universe does not care what you find elegant. The experiment has told you that your mental model needs to break and reform.

A language model will never give you that experience. It will give you Planck's answer. It will not give you Planck's confusion. And the confusion was where the physics happened.

If you skip that period, you do not arrive at the same destination faster. You arrive somewhere else entirely. You arrive at fluency. Fluency is not understanding. You can be fluent in the language of quantum mechanics without having any sense of where the weight-bearing walls are.

The Scientific Method, As It Was

The scientific method, as we typically teach it, runs in a cycle:

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observe → hypothesise → predict → experiment → revise
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Each step requires a different kind of thinking. Observation requires attention. Hypothesis requires imagination. Prediction requires mathematical rigour: you must derive a specific, falsifiable consequence from your model. Experiment requires honesty: you must accept the result even when it is not what you hoped. Revision requires humility: you must be willing to abandon an idea you liked.

The method is slow. It is meant to be slow. The slowness is not a bug. It is the mechanism by which false ideas are filtered out. Speed is the enemy of this process. Every shortcut that allows a plausible-sounding hypothesis to skip the experimental test is a corruption of the method.

The Scientific Method, As It Is Becoming

The temptation now is to let the tool handle the middle steps. You observe something. You ask the machine to hypothesise. It produces a model. You ask it to predict. It produces a formula. You skip the experiment because the formula looks right, and the prose surrounding it sounds authoritative.

This is not the scientific method. It is the scientific *aesthetic*. It has the shape of science without the substance. The substance is in the test. The test is the moment where the universe gets a vote. If you remove that vote, what remains is literature.

I say this with some self-awareness. I wrote a paper recently with the help of a language model that was a small piece of mathematics, not physics. But the reviewer's criticism, "you do not explain why this is important or what problem it solves," is the scientific method's question applied to a mathematical result: *so what?* What does this observation let you do that you could

not do before? If the answer is nothing, the observation is a curiosity. The machine produced a polished account of my curiosity. It could not tell me that the curiosity was not enough. That was literally my job. I was content to reveal my journey to an understanding that, to a learned eye, was of little consequence.

On Writing That Matters to the Reader

Larry McEnerney, who ran the University of Chicago writing programme for decades, makes this point better than I can. Your job as a writer is not to show what you know. It is to change what the reader thinks. If the reader's question is "so what?" and you have no answer, the writing has failed, no matter how polished it is.
Watch from around 11 minutes: `https://www.youtube.com/embed/vtIzMaLkCaM?start=655&list=PLK7HcR8L32zrOlnj1xxlIlFEUh3xYQ7Xp`

Two Disciplines for Your Writing

Because these tools produce polished language so readily, they introduce a particular temptation: your work can begin to sound finished before the thinking has finished or even started. Guard against this. The disciplines are simple.

In mathematics:

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claim → equation → consequence
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A claim should be tied to an explicit mathematical object. An equation should make the claim precise. A consequence should show what follows and why anyone should care. If you cannot state the claim without the machine's help, you do not yet understand the claim. If the equation does not appear naturally from the claim, the claim may be hollow.

In physics:

```
observation → model → prediction → measurement
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An observation should be tied to something you have actually seen or measured. A model should be the simplest structure that accounts for it. A prediction should be a specific number or relation that the model produces and that you did not put in by hand. And a measurement should be something you or someone else can actually perform. If the prediction cannot be tested, it is not yet physics. It is a sketch.

Strip everything back to these skeletons before you show your work to anyone. If the skeleton stands, you have something. If it does not, no amount of polished prose will save it.

On Honest Disclosure

When I use these wonderful LLM tools in my own work, I say so. Not dramatically. Not evasively. I may think of them as my research assistants, always keen and willing to back-fill my ideas and join some dots, but they do not have agency or an ability to appraise my callings. So in the

same spirit as a physicist who writes "numerical simulations were performed using MATLAB" or a mathematician who writes "symbolic computations were performed using Mathematica," the sentence I shall always use is:

> Expository structuring was assisted by Claude (Anthropic) / ChatGPT (OpenAI).
> The mathematical content and direction of argument are the author's.

This is, I hope, an honest account of our joint endeavours. It does not pretend the tool was absent. But it also does not pretend the tool had mathematical or physical judgment. It describes a workflow. You should do the same whenever you use these tools. Not because you are required to, but because the habit of honesty about your own process is itself a form of intellectual discipline. If you cannot say clearly which parts are yours and which parts were generated, that is a signal that you do not yet understand the work well enough to put your name on it.

What I Am Asking of You

I am not asking you to avoid these tools. They exist. They are useful. I use them myself. I am asking you to understand what they are and what they are not.

They are generators of candidate structures. They are not judges of truth. They are combinatorial assistants. They do not have comprehension. They do not take responsibility for claims. They cannot reject incorrect reasoning on mathematical grounds. They cannot run an experiment. They cannot tell you when a model contradicts nature. That is your job.

And to do that job, you must first build the internal machinery. There is no shortcut. The confusion is not an obstacle to understanding. The confusion *is* the process of understanding. Sit with it. Endure it. Let your mental structures break and reform. That is how the ladder gets climbed.

In mathematics, the struggle is with the proof that will not close, the conjecture that turns out to be false, the definition that needs to be rewritten for the third time. In physics, the struggle is with the experiment that gives the wrong answer, the model that is beautiful but predicts a number that is off by a factor of two, the graph that does not look like the theory said it should. Both struggles are necessary. Both build the same thing: the capacity to know, when you see a result, whether it is real.

The tools will get better. They will produce more convincing mathematics, more plausible physics, more fluent exposition. The gap between "generated" and "understood" will become harder to see from the outside. But it will remain. It will always remain. Because understanding is not a property of text. It is a property of the person who can stand behind the text, defend it, extend it, and know when it is wrong. In mathematics, that means knowing when a proof has a hole. In physics, that means knowing when a prediction needs to be tested against the world rather than accepted because it sounds right. That is what I want you to become. The tool cannot want it for you.

Lee McCulloch-James, PhD
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